Nexus B2b — Team Work Plan

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B2b gives Nexus a real AI agent that writes code inside jobs, so software engineering runs end to end. Six of its 18 tasks are done or nearly done; the remaining 12 are split across three members below.

Where B2b stands

Code lives on branch `our-platform` at github.com/Mohammedjaasir/Nexus (also pushed as `main`). The plan is `.superpowers/sdd/2026-10-21-nexus-b2b/b2b-plan.md` (Revision 1, 18 tasks); binding decisions are in `b2b-user-decisions.md` next to it.

Task	What	State
0	CLI feasibility probe and test harness	Done (efa76c41)
1	Shared contracts, recipes, registry, skills	Done (bf40686a), review fix pending: SDE must not be marked "end to end" yet
2	Schema v10 module	Done (155d6afa)
3	v10 migration step (opt-in)	In progress (Mohammed + Claude)
4	Tokens, agent authorization, consent, budgets	In progress (Mohammed + Claude)
7	Repository writes: approval, worktrees, git engine jobs	In progress (Mohammed + Claude)
12	Host egress proxy (api.anthropic.com only)	In progress (Mohammed + Claude)
5, 6, 8–11, 13–17	The rest of B2b	Not started — assigned below

Who does what

Each member owns one layer of the app, so no two people edit the same files. Put your name next to your member number.

Member	Area	Tasks	Main folders
Member 1 (name: ____)	Main process: agent sessions and change sets	5, 8, 9, 10	`src/main/nexus/agents`, `src/main/nexus/{sessions,review,repo,validation}`, `src/main/nexus/exec`
Member 2 (name: ____)	Runner, sandbox and the CLI adapter	6, 11, 13	`src/runner`, `tools`, `test/helpers` agent harness
Member 3 (name: ____)	IPC, screens, end-to-end tests and docs	14, 15, 16, 17	`src/main/nexusIpc`, `src/preload`, `src/renderer/src/nexus`, `docs/nexus`
Mohammed (with Claude)	Finishing tasks in progress; reviews and merges	3, 4, 7, 12	as in the plan

Order of work

Start a task only when the tasks it waits on are merged. Members can work in parallel inside the same wave.

Wave	Task	Owner	Waits on
A	5 MCP gateway, context, skills (main)	Member 1	4
A	8 Change sets: register, validators, lineage	Member 1	7
A	14 IPC skeleton for channels #42–#51 (views only)	Member 3	1
B	6 Loopback MCP server in the runner	Member 2	5
B	9 Review, accept with rollback, merge accepted work	Member 1	8

Wave	Task	Owner	Waits on
B	10 Agent turns in main: sessions, usage caps, recovery	Member 1	4, 5, 7
C	11 Agent turns in the runner, ScriptedRuntime, process trees	Member 2	6, 10
C	14 IPC wired to real services, diff preview	Member 3	8, 9, 10
D	13 CLI adapter and `test:runtime` gate	Member 2	11, 12
D	15 Screens: agent panel, change-set review, setup rows	Member 3	14
E	16 End-to-end flows and the real-Docker agent suite	Member 3 with Member 2	all above
E	17 Docs Part IV	Member 3	16

Member 1 — main process

You build how Nexus runs an agent turn and turns its work into a reviewable change. Read Tasks 5, 8, 9 and 10 in the plan before starting.

- [ ] **Task 5 — MCP gateway.** Every agent tool call goes through main: check the capability token, the tool, the project scope and the call cap, then audit it. Write the context file and skills main hands to the agent each turn.
- [ ] **Task 8 — Change sets.** Nexus (not the agent) commits the agent's work. Checks before a change set can be reviewed: non-empty diff, files only in scope, no provider keys or private keys (other random-looking strings only warn), tests pass at the head commit, merges cleanly. Ignore or refuse anything under `.claude/`.
- [ ] **Task 9 — Review, accept, merge.** Accept writes a durable marker, moves `refs/nexus/accepted/<branch>` by compare-and-set and rolls back on failure. "Merge accepted work" is a dialog-confirmed fast-forward to the confirmed commit and refuses a dirty tree. Accept never touches the user's checked-out branch.
- [ ] **Task 10 — Agent turns in main.** Sessions, scheduling guards (runtime, consent, budget, rate), usage caps enforced during the turn ($2/turn, $10/day, 6 turns/hour, 1 at a time), turns with no change end as `partial`, and restart recovery.

Done means: the task's tests pass, `npm run typecheck`, `npm run test:db` and `npm run test:focused` show only the 11 known Windows failures, and a reviewer approves.

Member 2 — runner, sandbox and CLI adapter

You build the side that actually runs the agent: the runner process, its local tool server and the connection to the Claude CLI. Read Tasks 6, 11 and 13 and the Task 0 report (`task-0-report.md`) first.

- [ ] **Task 6 — Loopback MCP server in the runner.** Listens on 127.0.0.1 only, checks Host and Origin, needs the bearer token, answers `server/discover` (or method-not-found) before `initialize`. Command output goes to the agent; main only gets sizes and a hash.
- [ ] **Task 11 — Agent turns in the runner.** The turn container (no network), the in-process `ScriptedRuntime` for tests, clean-up of the CLI's process tree on Windows within the same app run, and shutdown on quit.
- [ ] **Task 13 — CLI adapter and `test:runtime`.** Run the CLI with the pinned flags: `--restricted`, `--setting-sources ""`, `--strict-mcp-config`, `--permission-mode` with `--allowedTools`, and `--disallowedTools Task Agent PowerShell Bash WebFetch WebSearch NotebookEdit`. Do **not** use `--safe-mode` (it switches off Nexus's tools). Context goes in a file, never on the command line. Network only through the Task 12 proxy to api.anthropic.com:443.

Safety rule for tests: run the real CLI only inside the harness (fake home folder, fake key, mock model on localhost, deny-all proxy). Never against anyone's real login or the internet. Real-login checks are done once by Mohammed, at verification.

Member 3 — IPC, screens, tests and docs

You connect the new features to the app's screens and prove the whole flow works. Read Tasks 14–17, and match the existing Spine + Folio style (no dashboards or card grids).

- [ ] **Task 14 — IPC channels #42–#51.** Each goes through the existing `handler()` (availability, validation, user check, one service call). Views never carry tokens, host paths or secrets. Includes a change-set diff preview capped at 1 MiB per side, and channel #51 for agent consent.
- [ ] **Task 15 — Screens.** Agent panel, change-set review in the Changes tool, setup rows (the engine name and api.anthropic.com appear only there), and the intent line. Approvals say exactly what will happen. Do not edit the other session's files: App.tsx, RouteView.tsx, uiStore.ts, nexus/start/*, WelcomePage.tsx, start.css, welcome.css, hairline/*.
- [ ] **Task 16 — End-to-end.** Flows with `ScriptedRuntime` (Node and Python fixtures), the gated real-Docker agent suite, and only then flip SDE to "end to end" in the registry. Needs Member 2.

- [ ] **Task 17 — Docs.** Part IV of `docs/nexus/platform-core.md`: how agent turns work, the limits, the known limitations and the runbook.

Rules every member follows

1. **One branch per task** off `our-platform` (e.g. `b2b/task-8`); open a pull request into `our-platform` on github.com/Mohammedjaasir/Nexus. One reviewer approves before merge. No force-push, no rebase of shared branches, never push to `main` directly.
2. **Evidence, not exit codes.** Nothing is marked done because a command exited 0 or a file exists. Completion needs checked, saved evidence linked to the job and version.
3. **No secrets anywhere visible:** not in screens, logs, saved files, activity, prompts or error messages. The agent's login is never read, copied or stored by Nexus.
4. **Network is off by default.** The agent reaches only api.anthropic.com:443, through the Nexus proxy, after the user grants it once.
5. **Never weaken a test** to make it pass. The only allowed failures are the 11 known Windows ones in `.superpowers/sdd/2026-10-08-nexus-domains/baseline-fails.txt`.
6. **Database changes are additive and opt-in.** Never run v10 against a real database; tests use temporary copies.
7. **No vendor names in the data model, jobs or workflows** — the agent engine is a replaceable `cli-agent`.
8. **Ask before** anything irreversible, any new network destination, or any change to another member's files.

Setting up a machine

1. `git clone -b our-platform https://github.com/Mohammedjaasir/Nexus.git`, then `npm ci` and `npm run build`.
2. Node 20 for `node --test test/` (Node 22 runs `test/` differently).
3. Docker Desktop running, then `node tools/nexus-prepare-images.cjs --build` for the three pinned images.
4. At least **30 GB free** (40 GB recommended) before `npm run test:exec`; it refuses to run below that.
5. Claude CLI installed (2.1.x). Tests never use your login; only the final real-login check does.
6. Run the app only with a scratch folder: `--user-data-dir=<empty folder>` and `NEXUS_ENABLE_V9=1`.

Start by reading `.superpowers/sdd/2026-10-21-nexus-b2b/HANDOFF.md`, then your tasks in `b2b-plan.md`.

After B2b

Each phase is planned, built, verified and then approved before the next one starts.

Phase	Delivers
B2c	QA end to end, and the loop SDE → QA → defect → fix → retest → release
B3	Data Analyst: datasets, cleaning, queries, charts, reports
B4	Data Scientist: notebooks, statistics, experiments, baseline models
B5	ML Engineer: training, evaluation, model versions, local serving
B6	DL Engineer: CPU training, ONNX export, optimisation, serving